

6

MEASUREMENT AND SPACE

GEOMETRY AND NETWORKS

During the 18th century, in the Prussian town of Königsberg (now Kaliningrad, Russia, pictured), there were 7 bridges connecting 2 islands and the banks of the Pregel River (see map next page). There was a tradition on Sundays where people would walk through the town try to complete a walk crossing each bridge only once. No person was able to do it but no one could prove that it was impossible.

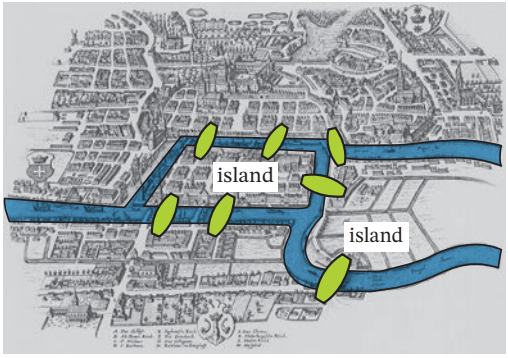
In 1736, the Swiss mathematician Leonhard Euler solved the problem by reducing the map to a network diagram, creating a new field of mathematics called **graph theory**. Euler proved that it was impossible to do the walk. Eventually, another bridge was built.


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Chapter outline

	Working mathematically				
6.01 Triangle geometry	U	F		R	C
6.02 Quadrilateral geometry	U	F		R	C
6.03 Networks*	U	F		R	C
6.04 Planar graphs and Euler's formula*	U	F		R	C

*PATH



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Wordbank

angle sum The total of the sizes of all the angles in a shape, such as a triangle

bisect To cut in half

convex quadrilateral A quadrilateral whose vertices all point outwards

diagonal An interval joining 2 non-adjacent vertices of a shape

edge (of a network) A line that connects vertices (points) together

exterior angle of a triangle An 'outside' angle of a triangle formed after extending one of the sides of the triangle

planar graph A connected network (graph) where the edges do not cross (overlap)

vertex (of a network) A point representing an item that is connected by lines called edges.

Quiz
Wordbank 6

U = Understanding | **F** = Fluency | **PS** = Problem solving | **R** = Reasoning | **C** = Communicating

Videos (5):

6.01 Angle sum of a triangle • Exterior angle of a triangle 1 • Exterior angle of a triangle 2

6.03 Networks

6.04 Euler's formula for planar graphs

***Twig* videos (2):**

6.03 The seven bridges of Konigsberg • Networks: Labyrinths and mazes

Quizzes (5):

- Wordbank 6
- SkillCheck 6
- Mental skills 6
- Language of maths 6
- Test yourself 6

Skillsheet (1):

6.01 Naming shapes

Worksheets (6):

6.02 Classifying quadrilaterals • Properties of triangles and quadrilaterals • Deductive geometry • Diagonal properties of quadrilaterals • Shapes and angles review

Mind map Geometry and networks

Puzzles (3):

6.02 Mixed angle problems • Parallel lines • Triangles and quadrilaterals

Presentation (1):

6.02 Angles and shapes

Nelson MindTap

To access resources above, visit **cengage.com.au/nelsonmindtap**

In this chapter you will:

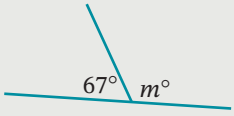
- classify triangles and quadrilaterals
- solve geometrical problems involving the angle sum of a triangle and quadrilateral and the exterior angle of a triangle
- (PATH) describe networks using appropriate terminology, including vertex, edge, path and degree
- (PATH) describe a planar graph and apply Euler's formula

Quiz
SkillCheck 6

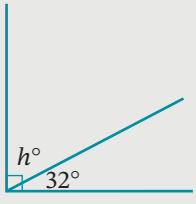
SkillCheck ANSWERS ON P. 575

1 Find the value of each variable.

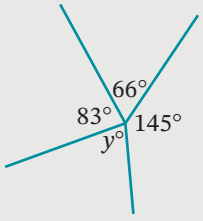
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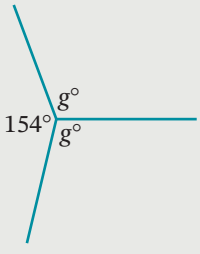
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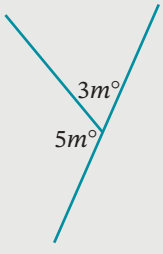
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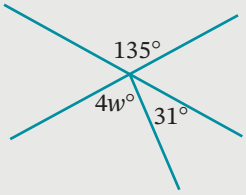
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f



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2 Find the complement of each angle size.

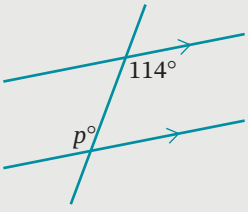
a 72° **b** 28°

3 Find the supplement of each angle size.

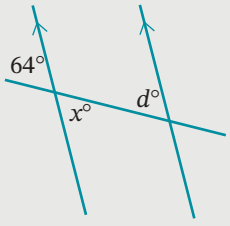
a 119° **b** 45°

4 Find the value of each variable, giving reasons.

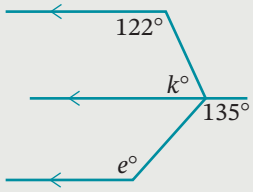
a



b

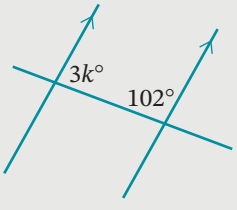


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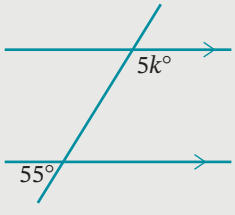


5 Find the value of k.

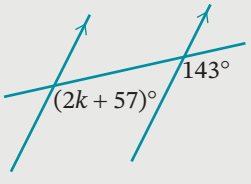
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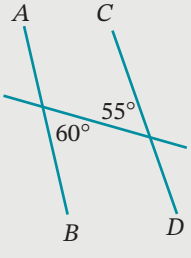


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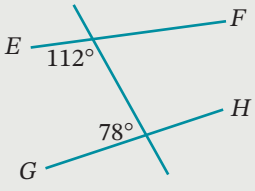


6 Test whether each labelled pair of lines are parallel. Give a reason for each answer.

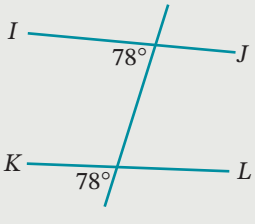
a



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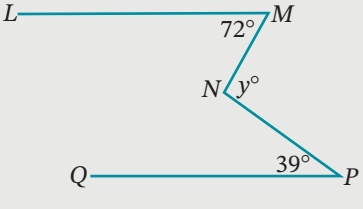


c



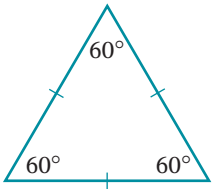
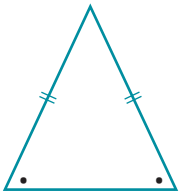
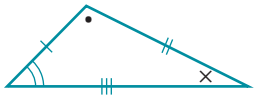
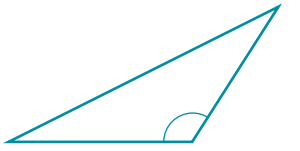
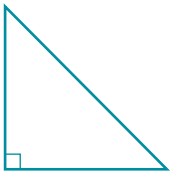
7 Find the value of y, giving reasons.

a



6.01 Triangle geometry

Triangles can be classified in 2 ways: by their sides or by their angles.

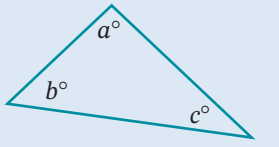
Classifying by sides		
Equilateral triangle	**Isosceles triangle**	**Scalene triangle**
		
3 equal sides (Also 3 equal angles, each 60°)	2 equal sides (Also 2 equal angles, opposite the equal sides)	No equal sides (Also no equal angles)
Classifying by angles		
Acute-angled triangle	**Obtuse-angled triangle**	**Right-angled triangle**
		
3 acute angles (less than 90°)	One obtuse angle (between 90° and 180°)	One right angle (90°)

Video
Angle sum of a triangle

Angle sum of a triangle

The **angle sum of a triangle** is 180°.

$a + b + c = 180$



Proof:

Consider any triangle PQR with angles $a°$, $b°$ and $c°$.

Construct a line through P parallel to QR.

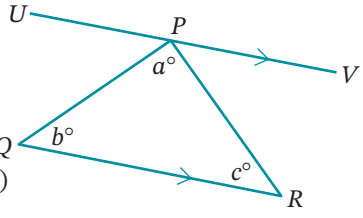
$\therefore \angle UPQ = b°$ (alternate angles, $UV \parallel QR$)

and $\angle VPR = c°$ (alternate angles, $UV \parallel QR$)

But $\angle UPQ + \angle QPR + \angle VPR = 180°$ (angles on a straight line)

$\therefore a° + b° + c° = 180°$

$\therefore$ The sum of the angles of a triangle is 180°.

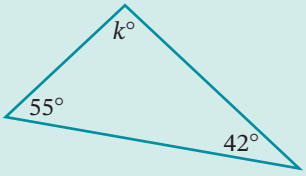


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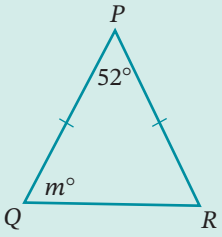
Example 1

Find the value of each variable, giving reasons.

a



b



Solution

a $k + 55 + 42 = 180$ (angle sum of a triangle)

$k = 180 - 55 - 42$

$= 83$

b $\angle R = \angle Q = m°$ ($\triangle PQR$ is isosceles)

$m + m + 52 = 180$ (angle sum of a triangle)

$2m + 52 = 180$

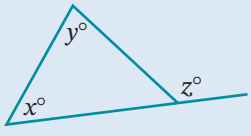
$2m = 128$

$m = 64$

The exterior angle of a triangle

The **exterior angle of a triangle** is equal to the sum of the 2 interior opposite angles.

$z = x + y$

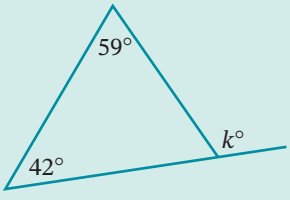


Video
Exterior angle of a triangle 1

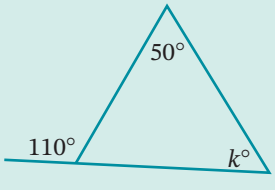
Example 2

Find the value of k in each diagram.

a



b



Video
Exterior angle of a triangle 2

Solution

a $k = 42 + 59$ (exterior angle of triangle)

$= 101$

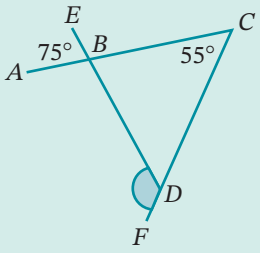
b $k + 50 = 110$ (exterior angle of triangle)

$k = 60$

Video
Exterior angle of a triangle 2

Example 3

Find the size of the marked angle $\angle BDF$.



Solution

$\angle DBC = \angle ABE = 75°$ (vertically opposite angles)

$\angle BDF = 75° + 55°$ (exterior angle of $\triangle BCD$)

$= 130°$

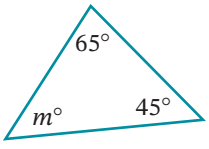
EXERCISE 6.01 ANSWERS ON P. 575

Triangle geometry U F R C

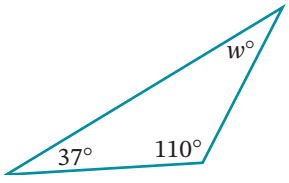
EXAMPLE 1

1 Find the value of each variable, giving reasons. R C

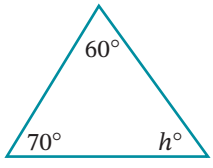
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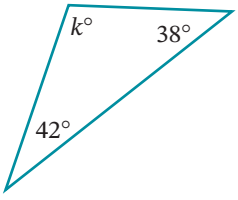
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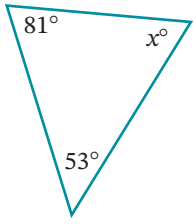
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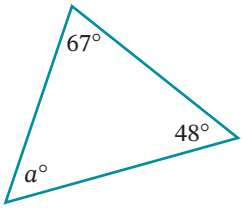
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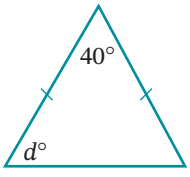
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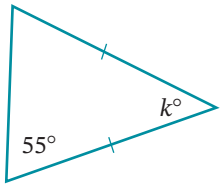
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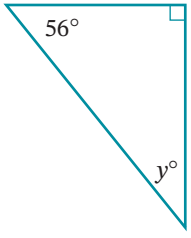
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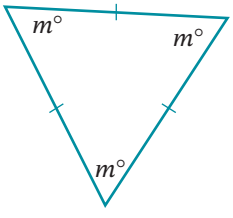
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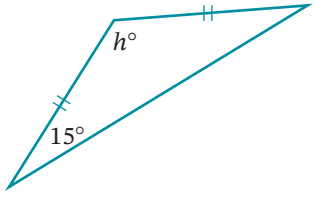
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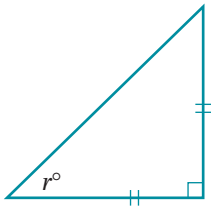
j



k



l



Foundation Standard Complex

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2 What is the size of each angle in an equilateral triangle? Select the correct answer **A**, **B**, **C** or **D**.

A 30° **B** 45° **C** 60° **D** 90°

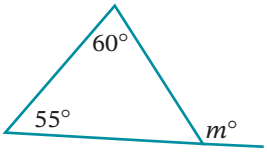
3 One angle of an isosceles triangle is equal to 80°. Which 2 of the following could be the sizes of the other angles? Select 2 of **A**, **B**, **C** or **D**. R

A 80° and 20° **B** 40° and 60° **C** 40° and 40° **D** 50° and 50°

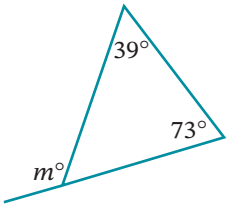
EXAMPLE 2

4 Find the value of m. R C

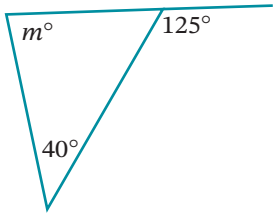
a



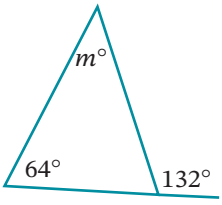
b



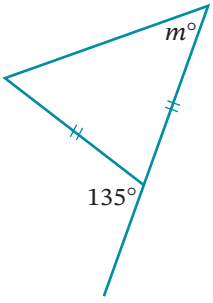
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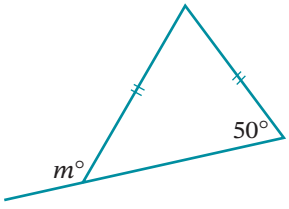
d



e



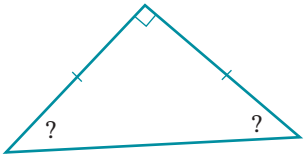
f



5 What is the size of the unknown angles in this triangle? Select **A**, **B**, **C** or **D**. R

A 30° **B** 45°

C 60° **D** 90°



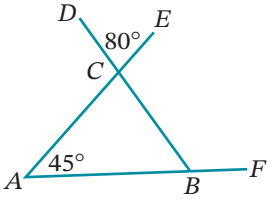
6 One angle of an obtuse-angled triangle is 50°. Which of the following could be the sizes of the other angles? Select **A**, **B**, **C** or **D**. R

A 80° and 50° **B** 100° and 30° **C** 65° and 65° **D** 45° and 95°

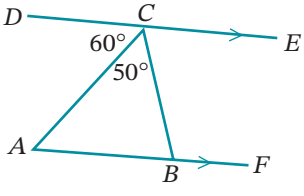
EXAMPLE 3

7 Find the size of $\angle CBF$ in each diagram. R C

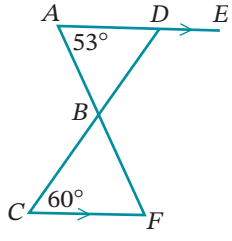
a



b

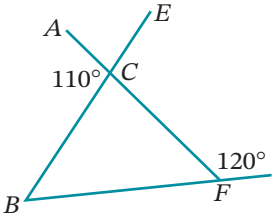


c

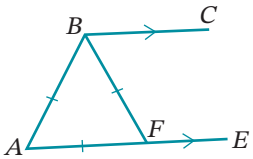


Foundation Standard Complex

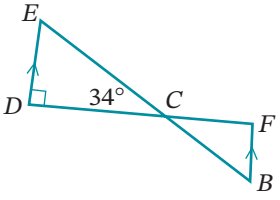
d



e

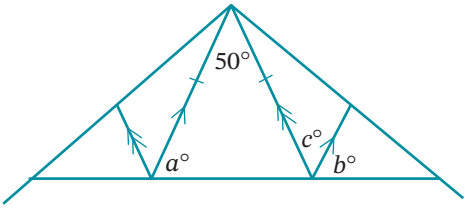


f

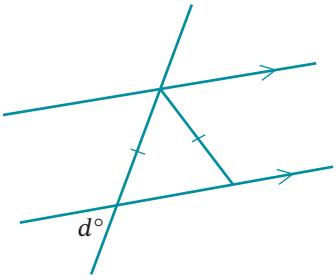


8 An isosceles triangle has a side length of 6 cm and one of its angles equal to 40°. Draw all possible shapes of the triangle. R

9 The diagram shows the shape of a roof truss. Find the value of each variable. R

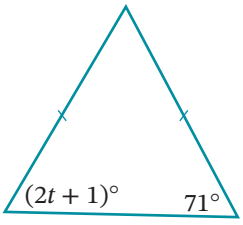


10 Copy this diagram and mark all angles equal to the angle marked $d°$. R

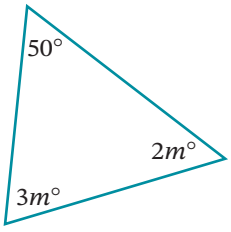


11 Find the value of each variable. R C

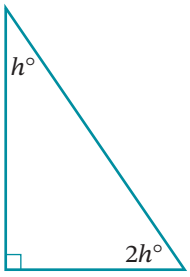
a



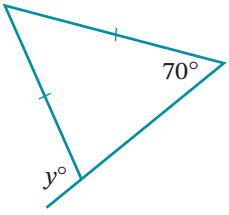
b



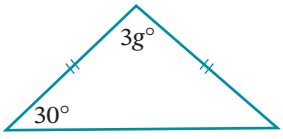
c



d



e



f



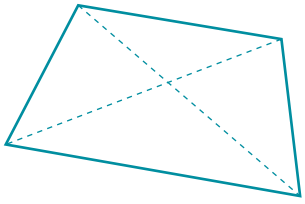
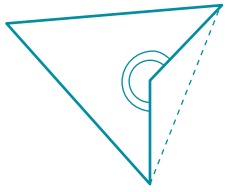
Foundation Standard Complex

9780170479370

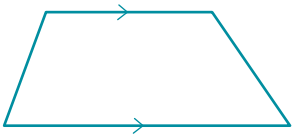
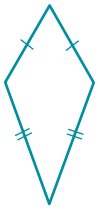
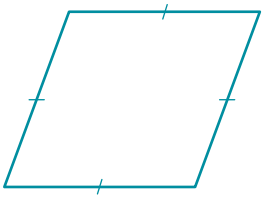
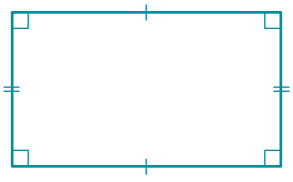
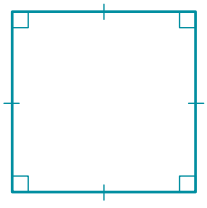
Quadrilateral geometry

6.02

A **quadrilateral** is any shape with 4 straight sides. A quadrilateral may be either **convex** or **non-convex** (**concave**).

Convex quadrilateral	Non-convex quadrilateral
	
• All **vertices** (corners) point outwards. • All **diagonals** lie within the shape. • All angles are less than 180°.	• One vertex points inwards. • One diagonal lies outside the shape. • One angle is more than 180° (reflex angle).

There are 6 special types of quadrilaterals.

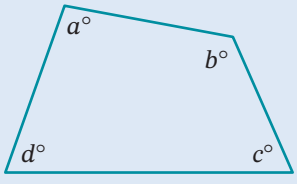
Trapezium	Kite	Parallelogram
		
One pair of parallel sides	2 pairs of equal adjacent sides 'Adjacent' means 'next to each other'.	2 pairs of parallel sides
Rhombus	**Rectangle**	**Square**
		
4 equal sides	4 right angles	4 equal sides and 4 right angles

Angle sum of a quadrilateral

The **angle sum of a quadrilateral** is 360°.

$a + b + c + d = 360$

This property is true for both convex and non-convex quadrilaterals.



Worksheets
Classifying quadrilaterals
Properties of triangles and quadrilaterals
Deductive geometry
Diagonal properties of quadrilaterals
Shapes and angles review

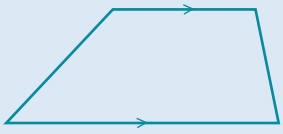
Skillsheet
Naming shapes

Puzzles
Mixed angle problems
Parallel lines
Triangles and quadrilaterals

Presentation
Angles and shapes

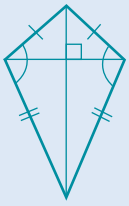
Properties of quadrilaterals

Trapezium



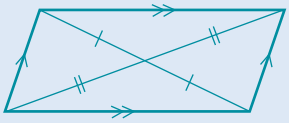
- One pair of opposite sides parallel
- No axes of symmetry

Kite



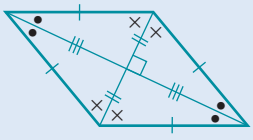
- 2 pairs of equal adjacent sides
- One pair of opposite angles equal
- One axis of symmetry
- Diagonals intersect at right angles

Parallelogram



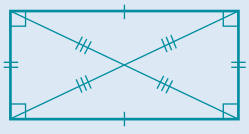
- Opposite sides are parallel and equal
- Opposite angles are equal
- No axes of symmetry
- Diagonals bisect each other

Rhombus



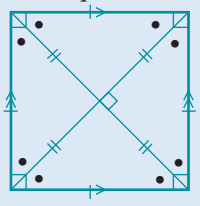
- All sides are equal
- 2 axes of symmetry
- A special type of parallelogram
- Diagonals bisect each other at right angles
- Diagonals bisect the angles of the rhombus

Rectangle



- All angles are 90° (right angles)
- 2 axes of symmetry
- A special type of parallelogram
- Diagonals are equal (in length)
- Diagonals bisect each other

Square

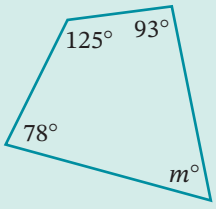


- All sides are equal
- All angles are 90° (right angles)
- 4 axes of symmetry
- A special type of rhombus and rectangle
- Diagonals are equal
- Diagonals bisect each other at right angles
- Diagonals bisect the angles of the square

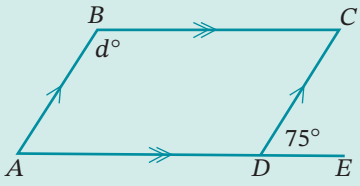
Example 4

Find the value of each variable.

a



b



Solution

a $m + 78 + 125 + 93 = 360$ (angle sum of a quadrilateral)

$m = 360 - 78 - 125 - 93$

$= 64$

b $\angle CDA = 180° - 75°$ (angles on a straight line)

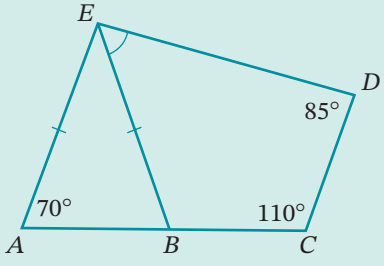
$= 105°$

$d = 105$ (opposite angles of a parallelogram)

Can you see another method for finding d?

Example 5

Find the size of $\angle BED$.



Solution

$\angle ABE = 70°$ (equal angles in isosceles $\triangle ABE$)

$\angle EBC = 180° - 70°$ (angles on a straight line)

$= 110°$

$\angle BED = 360° - 85° - 110° - 110°$ (angle sum of quadrilateral $BCDE$)

$= 55°$

Can you see another method for finding $\angle BED$?

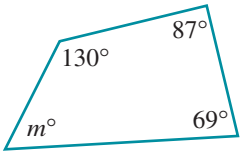
EXERCISE 6.02 ANSWERS ON P. 575

Quadrilateral geometry U F R C

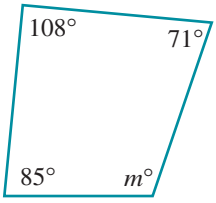
EXAMPLE 4

1 Find the value of m in each diagram. R

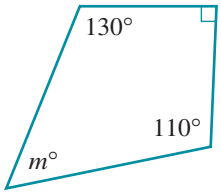
a



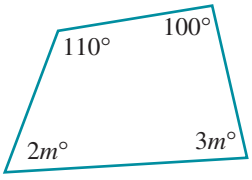
b



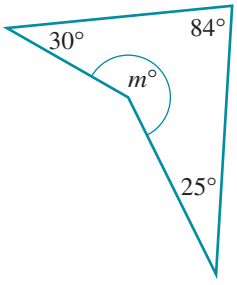
c



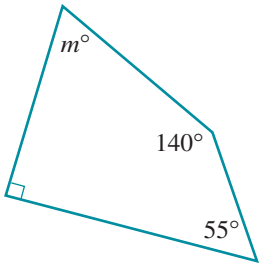
d



e



f



2 Copy and complete this table. R C

Property	Trapezium	Kite	Parallelogram	Rhombus	Rectangle	Square
One pair of opposite sides parallel						
Opposite sides parallel						
Opposite sides equal			✓			
All sides equal						
2 pairs of adjacent sides equal						
Diagonals equal					✓	
Diagonals bisect each other						
Diagonals meet at right angles						
Diagonals bisect the angles of the shape						
Opposite angles equal						
One pair of opposite angles equal						
All angles 90°						
Axes of symmetry	0					

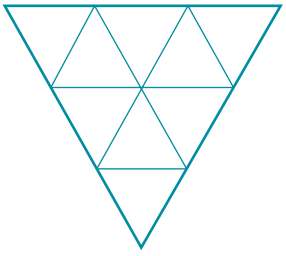
Foundation Standard Complex

9780170479370

3 Refer to the table you completed in question **2**, and name all quadrilaterals that have: R C

a no axes of symmetry
b one pair of parallel sides
c 4 equal sides
d equal diagonals
e opposite sides equal
f 4 axes of symmetry
g adjacent sides equal
h one axis of symmetry
i opposite sides parallel
j all angles measuring 90°
k 2 axes of symmetry
l diagonals which bisect each other
m opposite angles equal
n diagonals meeting at 90°

4 Each triangle in this diagram is an equilateral triangle. R



a Name the different types of quadrilaterals that you can find in the diagram.
b How many of each type of quadrilateral are there in the diagram?
c How many triangles are there? (It may help to copy the diagram so that you can draw on it using coloured pencils.)

5 Which statements are *always* true? R

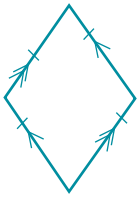
A A rhombus is a parallelogram.
B The diagonals of a parallelogram meet at right angles.
C A square is a rhombus.
D A parallelogram is a quadrilateral with a pair of opposite sides equal and parallel.
E A square is a rectangle.
F The diagonals of an isosceles trapezium bisect each other.
G The opposite angles of a rhombus are equal.
H The diagonals of a rhombus are equal and bisect each other at right angles.
I A rectangle is a parallelogram.

6 Name each quadrilateral using the properties marked on it. R

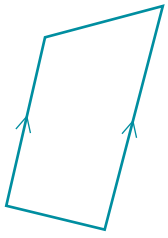
a



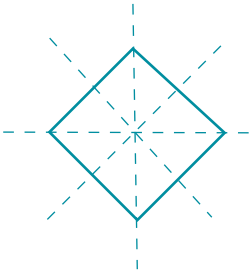
b



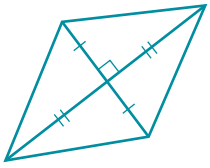
c



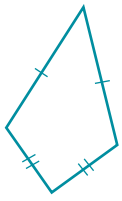
d



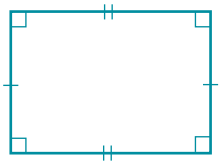
e



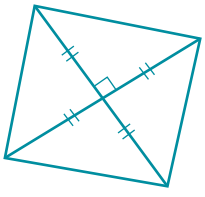
f



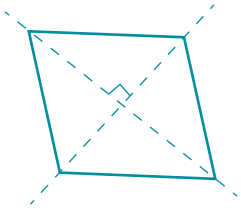
g



h

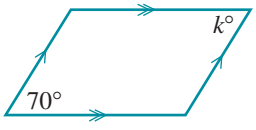


i

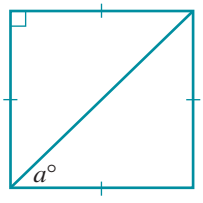


7 Find the value of each variable, giving reasons. R C

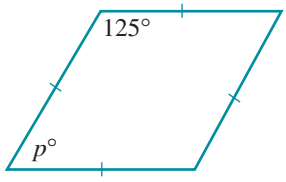
a



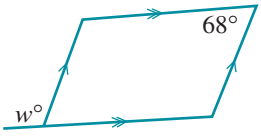
b



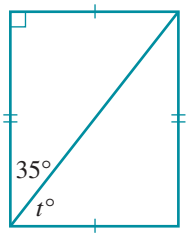
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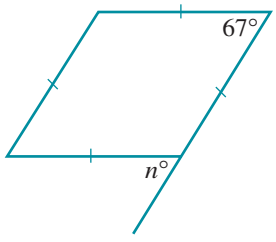
d



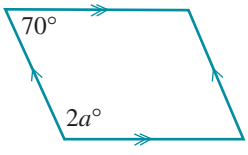
e



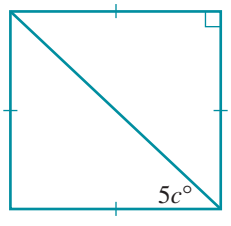
f



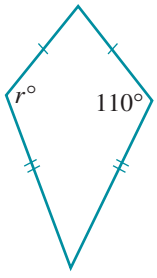
g



h



i



Foundation Standard Complex

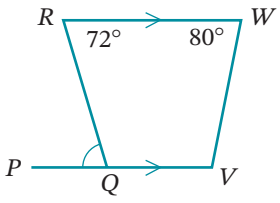
9780170479370

EXAMPLE 5

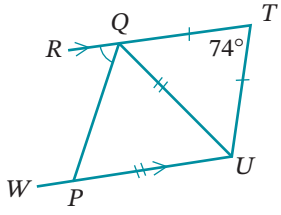
6.02

8 Find the size of $\angle PQR$ in each diagram, giving reasons. R C

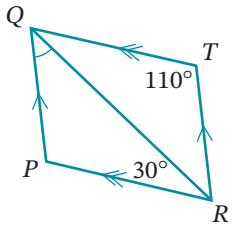
a



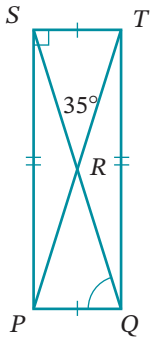
b



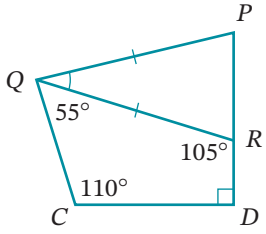
c



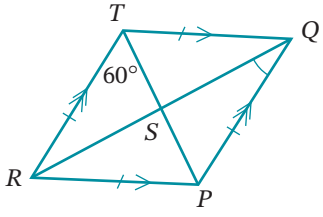
d



e



f



9 Use geometrical instruments or dynamic geometry software to construct each quadrilateral.

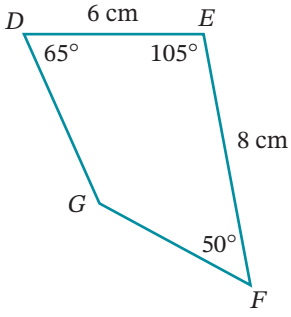
a Parallelogram $TUVW$ with sides $TU = 30$ mm, $UV = 60$ mm and $\angle TUV = 113°$.

b Rhombus with side length 5 cm.

c Convex quadrilateral $WXYZ$ with $WX = 65$ mm, $WZ = 40$ mm, $\angle ZWX = 54°$, $\angle WZY = 114°$ and $XY = 24$ mm.

d Non-convex quadrilateral with one side 4.5 cm and one angle 200°.

e



Foundation Standard Complex

Did you know?

Geometry in art

Geometry has been used in artwork for centuries.

Tessellations can be used to create artwork, such as the regular hexagons used in the tessellation shown.

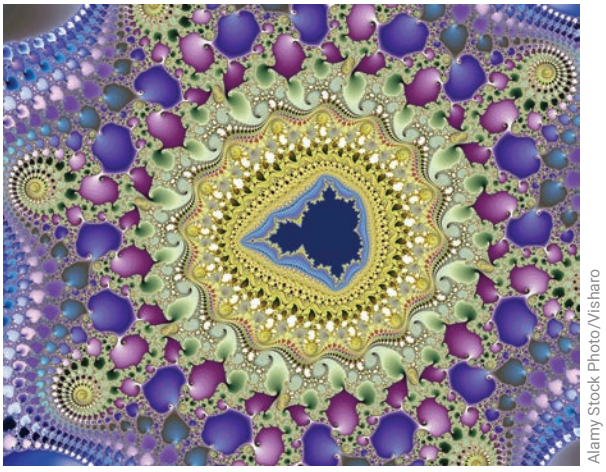


Indian and Islamic artworks show incredible intricacy, detail and colour within the geometric images used.



iStock.com/sharrocks

Fractals use geometric formulas to create infinitely occurring images, so they are usually created using computer software. A famous fractal is the Mandelbrot set, shown here.



Alamy Stock Photo/Visharo

1. Design a tessellation or find examples of geometry in art.
2. Investigate fractals such the Mandelbrot set and the Koch snowflake, and their history.

Mental skills 6: Maths without calculators ANSWERS ON P. 577

Quiz
Mental skills 6

Dividing decimals

To divide one decimal by another, first move the decimal points in both decimals the same number of places to the right so that the second decimal is a whole number.

1 Study each example.

a $0.24 \div 0.06 = 24 \div 6 = 4$

b $0.45 \div 0.5 = 4.5 \div 5 = 0.9$

c $0.006 \div 0.3 = 0.06 \div 3 = 0.02$

d $27 \div 0.9 = 270 \div 9 = 30$

e $1.6 \div 0.4 = 16 \div 4 = 4$

f $5.6 \div 0.07 = 560 \div 7 = 80$

2 Now evaluate each quotient.

a $0.25 \div 0.5$ **b** $63 \div 0.7$ **c** $3.2 \div 0.4$ **d** $0.18 \div 0.2$

e $2.7 \div 0.03$ **f** $0.042 \div 0.06$ **g** $4 \div 0.5$ **h** $1.2 \div 0.04$

i $0.072 \div 0.9$ **j** $0.35 \div 0.1$ **k** $0.28 \div 0.07$ **l** $0.033 \div 0.11$

3 Study each example.

Given that $112 \div 14 = 8$, evaluate each expression.

a
$$\begin{aligned} 112 \div 1.4 &= 112.0 \div 1.4 \\ &= 1120 \div 14 \\ &= 112 \times 10 \div 14 \\ &= 112 \div 14 \times 10 \\ &= 8 \times 10 \\ &= 80 \end{aligned}$$

Estimate: $112 \div 1.4 \approx 112 \div 1 = 112$

b
$$\begin{aligned} 0.112 \div 0.14 &= 0.112 \div 0.14 \\ &= 11.2 \div 14 \\ &= 112 \div 10 \div 14 \\ &= 112 \div 14 \div 10 \\ &= 8 \div 10 \\ &= 0.8 \end{aligned}$$

Estimate: $0.112 \div 0.14 \approx 0.1 \div 0.1 = 1$

c
$$\begin{aligned} 1120 \div 1.4 &= 11\,200 \div 14 \\ &= 112 \times 100 \div 14 \\ &= 112 \div 14 \times 100 \\ &= 8 \times 100 \\ &= 800 \end{aligned}$$

Estimate: $1120 \div 1.4 \approx 1120 \div 1 = 1120$

d
$$\begin{aligned} 1.12 \div 14 &= 112 \div 100 \div 14 \\ &= 112 \div 14 \div 100 \\ &= 8 \div 100 \\ &= 0.08 \end{aligned}$$

Estimate: $1.12 \div 14 \approx 1.12 \div 10 = 0.112$

4 Now given that $368 \div 23 = 16$, evaluate each quotient.

a $36.8 \div 2.3$ **b** $368 \div 2.3$ **c** $3.68 \div 2.3$ **d** $0.368 \div 0.23$

e $36.8 \div 23$ **f** $3.68 \div 0.23$ **g** $36.8 \div 0.23$ **h** $0.368 \div 2.3$

I $0.368 \div 23$ **j** $3.68 \div 0.023$ **k** $3.68 \div 23$ **l** $0.368 \div 230$

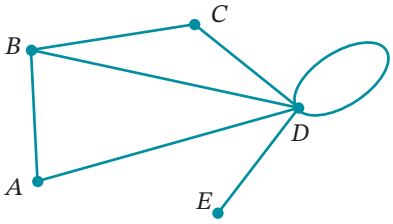
6.03 Networks

PATH

Videos
Networks
The seven bridges of Konigsberg
Networks: Labyrinths and mazes

A **network** is a collection of objects connected to each other in some way or system. Examples of networks include bus, rail and air routes, computer systems and family or friend relationships.

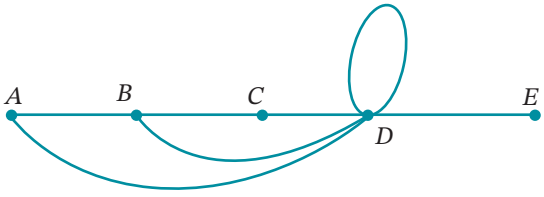
A **graph** is a diagram of a network where the objects are represented by points called **vertices** (or **nodes**), which are joined by lines called **edges**. The study of networks is called **graph theory**.



This graph or network diagram has 5 vertices, *A, B, C, D, E,* and 7 edges, *BC, BD, BA, AD, ED, CD, DD.*

A **loop** is an edge that connects a vertex to itself, such as *DD* in the diagram.

A network can be drawn in different ways as long as it shows the same connections. This diagram represents the same network as above.



Degree of a vertex

The **degree** (or **order**) of a vertex is the number of edges connected to it, with loops counted as 2 edges. An **even vertex** has an even number of edges connected to it. An **odd vertex** has an odd number of edges.

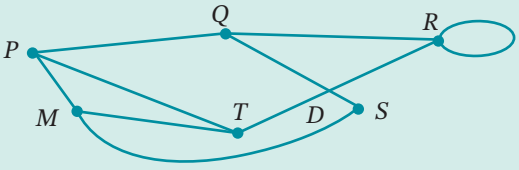
The **sum of the degrees** of all vertices in a network is equal to **twice the number of edges**, because each edge belongs to 2 vertices. This means that the sum of the degrees in a network is always even.

Example 6

a Find the degree of each vertex in this network.

b How many vertices are odd?

c Show that the sum of the degrees is equal to twice the number of edges.



Solution

a

Vertex	*M*	*P*	*Q*	*R*	*S*	*T*
Degree	3	3	3	4	2	3

b There are 4 odd vertices.

c Number of edges = 9.

Sum of degrees of vertices = 3 + 3 + 3 + 4 + 2 + 3 = 18,

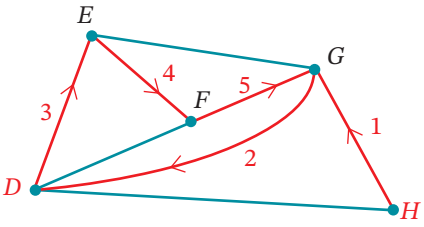
which is twice the number of edges (2 × 9).

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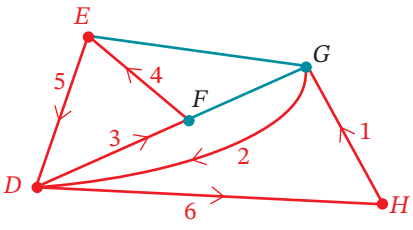
Walks, trails, paths and cycles

A **walk** is *any* route along the edges of a network, and may include repeated vertices or edges. Any walk that starts and ends at the same vertex is called a **closed walk**.

A **trail** is a walk with no repeated edges, such as shown in the network below. However, a vertex can be repeated, such as *G* in *HGDEFG*. A **circuit** is a **closed trail**, such as *HGDFEDH* shown below (starts and ends at *H*).



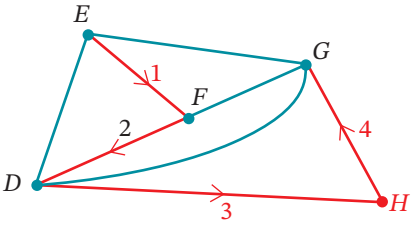
trail



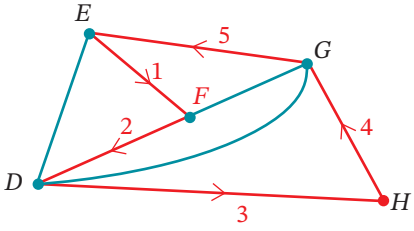
circuit

Think of a bushwalker walking on a trail or circuit and not repeating an edge.

A **path** is a walk with no repeated vertices (or edges), such as *EFDHG* shown in the network below. A **cycle** is a **closed path**, such as *EFDHGE* shown below (starts and ends at *E*, no repeated vertices).



path



cycle

Think of a cyclist riding along a bicycle path or cycle and not revisiting any points.

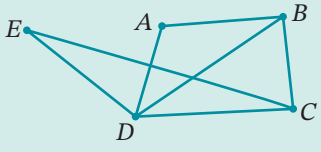
Example 7

a State whether each walk is a trail or a path.

i *BCED* **ii** *ADECBDC*

b Write down a circuit that uses the edge *DC*.

c Write down 2 cycles that use the edge *DC*.



Solution

a **i** *BCED* is a path since no vertices (and edges) are repeated.

ii *ADECBDC* is a trail as no edges are repeated but vertex *D* is repeated.

b *ABDCEDA* is a circuit because it starts and ends at *A* with no repeated edges.

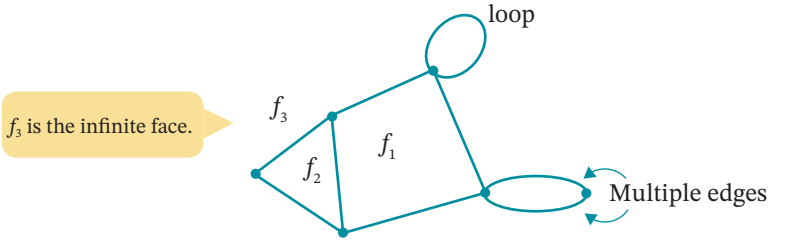
c *ADCBA* and *CEDC* are cycles because they have no repeated vertices.

PATH

Faces of a network

A **face** of a network is a region bounded by edges. Regions formed by loops or multiple edges between the same vertices are *not* faces, but the external (outside) region of a network is a face. The external face is called the **infinite face**.

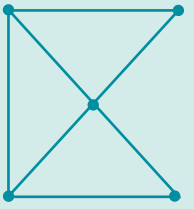
This network has 3 faces, f_1, f_2 and f_3.



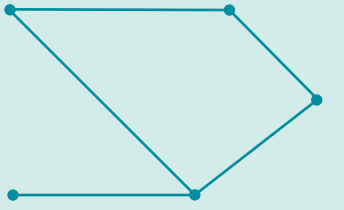
Example 8

Count the number of faces in each network.

a

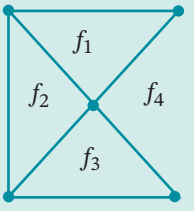


b



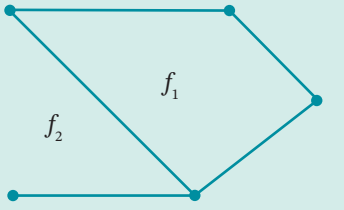
Solution

a



4 faces

b



2 faces

EXERCISE 6.03 ANSWERS ON P. 577

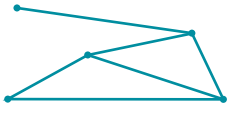
Networks UFRC

EXAMPLE 6

1 For each network, count the number of:

i vertices **ii** edges

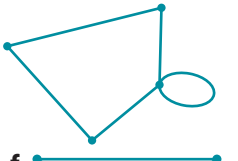
a



b



c



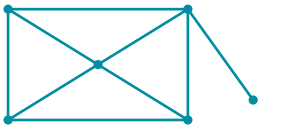
d



e



f

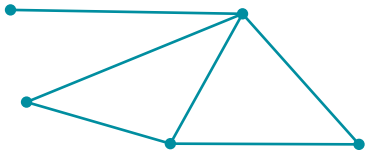


Foundation Standard Complex

9780170479370

2 How many even vertices has this network?

Select the correct answer **A**, **B**, **C** or **D**.

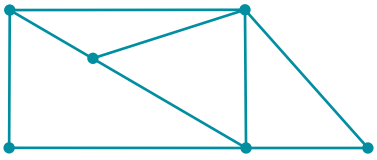


A 1 **B** 2

C 3 **D** 4

3 How many odd vertices has this network?

Select **A**, **B**, **C** or **D**.



A 2 **B** 3

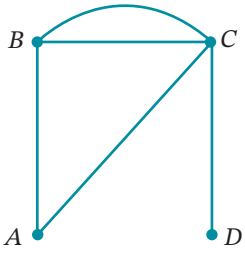
C 4 **D** 5

4 For each network: R C

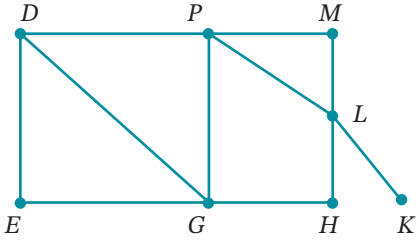
i find the degree of each vertex

ii show that the sum of the degrees is equal to twice the number of edges.

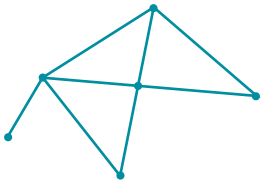
a



b



5 What is the sum of the degrees of all vertices in this graph? Select **A**, **B**, **C** or **D**.

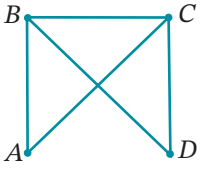


A 8 **B** 12

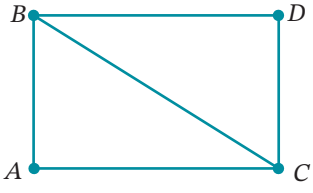
C 16 **D** 20

6 Which of these graphs does *not* show the same information as the given network diagram?

Select **A**, **B**, **C** or **D**. R C



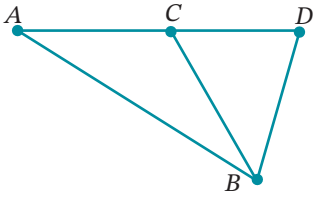
A



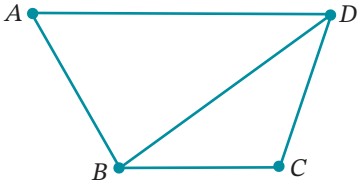
B



C

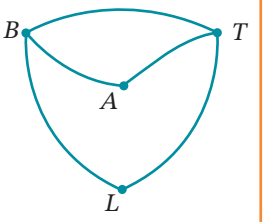


D



PATH

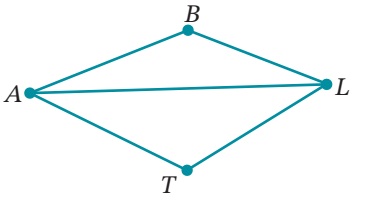
7 This network shows the friendship connections between Lara (L), Tasnuba (T), Aditi (A) and Belinda (B). An edge represents a friendship between 2 people.



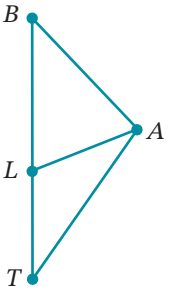
Which graph shows the same information?

Select **A**, **B**, **C** or **D**. R C

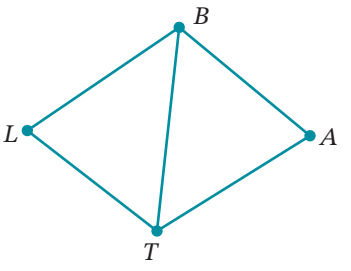
A



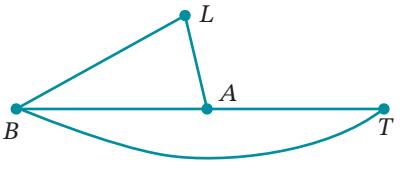
B

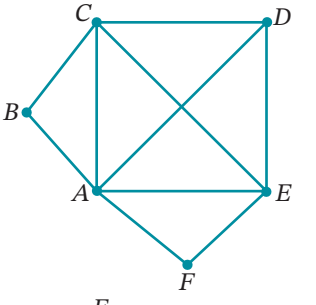


C



D



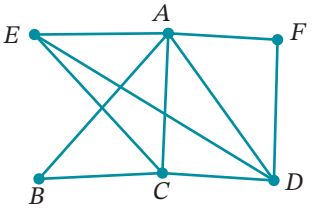


8 The graph shows the communication network between division managers in a business.

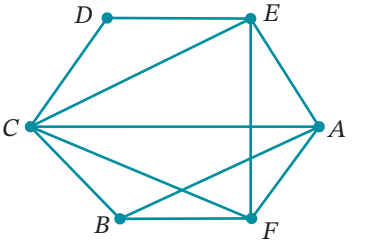
Which graph shows the same information?

Select **A**, **B**, **C** or **D**. R C

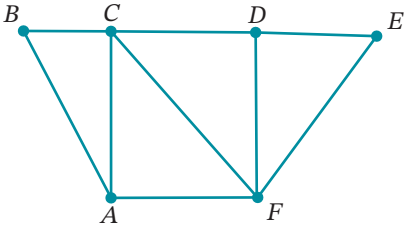
A



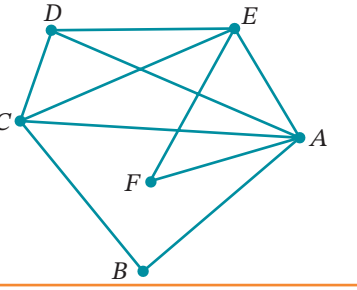
B



C



D



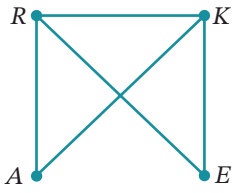
Foundation Standard Complex

9780170479370

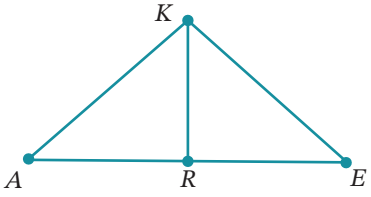
9 This graph shows social media connections between Ashley (A), Emily (E), Kristy (K) and Robyn (R).

Which graph shows the same information?

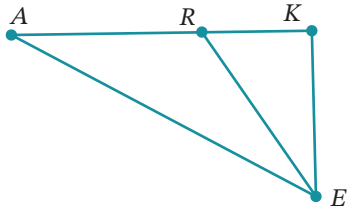
Select **A**, **B**, **C** or **D**. R C



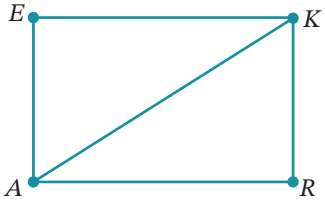
A



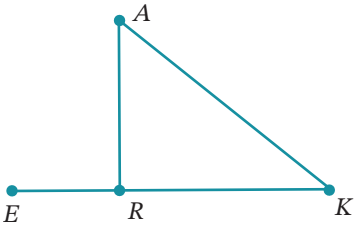
B



C

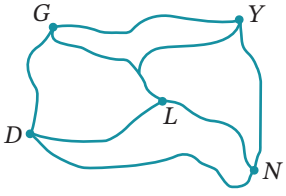


D

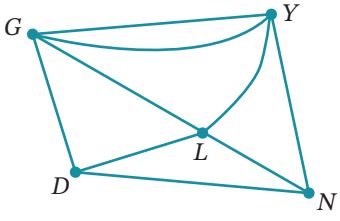


10 Which network diagram represents this map showing the roads that connect 5 towns?

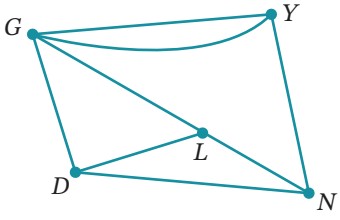
Select **A**, **B**, **C** or **D**. R C



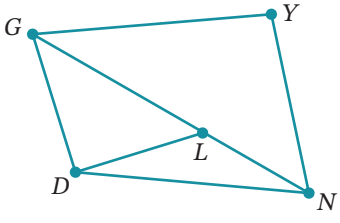
A



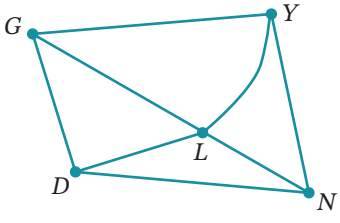
B



C

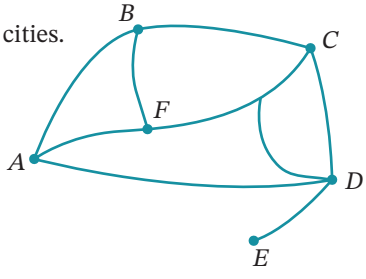


D



11 The diagram shows the airline connections between 6 cities.

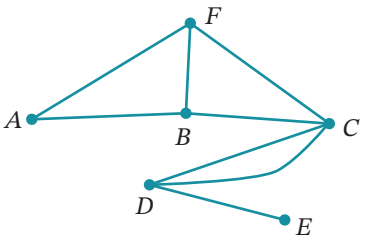
Which of the following represents the same information? Select **A**, **B**, **C** or **D**. R C



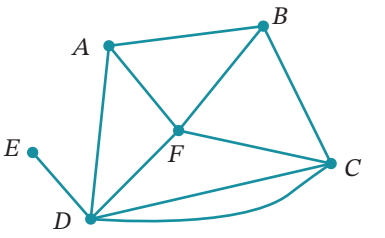
Foundation Standard Complex

PATH

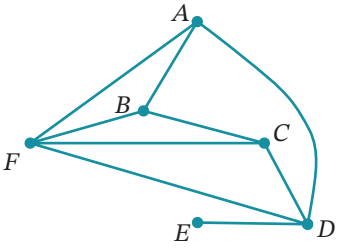
A



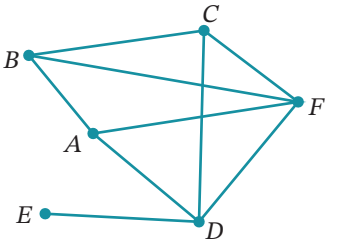
B



C



D



12 Arya, Cassie, Naved, Lachlan, Riya and Shayaan attend the same high school.

Naved is friends with Arya, Shayaan and Riya

Arya is friends with Naved and Riya

Shayaan is friends with Naved, Riya, Lachlan and Cassie

Lachlan is friends with Riya, Shayaan and Cassie

Cassie is friends with Shayaan and Lachlan

Riya is friends with Naved, Arya, Shayaan and Lachlan R C

a Draw a graph to represent the above network of friends.

b Who is not a friend with Riya?

EXAMPLE 7

13 **a** State whether each walk is a trail or a path. R C

i *ABDCED*

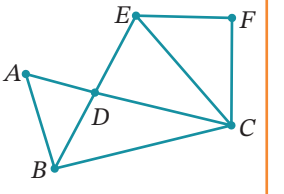
ii *CFEDBC*

iii *FEDBA*

iv *DECBA*

v *BDCFEDAB*

vi *BDCB*



b Which of the walks in part **a** is:

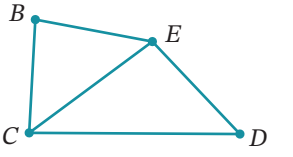
i a circuit (a closed trail)?

ii a cycle (a closed path)?

c Write down a circuit that contains the edge *DC*.

d Write down a cycle that contains the edges *AB* and *FE*.

14 Which of the following is a trail? Select **A**, **B**, **C** or **D**. R C



A *CEDCB*

B *DECBEC*

C *ECBDE*

D *CBECDEB*

Foundation Standard Complex

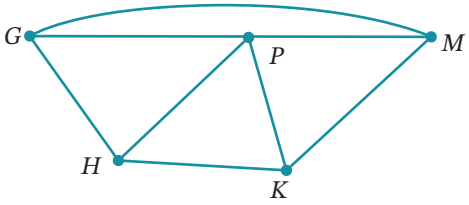
9780170479370

PATH

6.03

15 Which of the following is a circuit?

Select **A**, **B**, **C** or **D**. R C



A *HPKMPH*

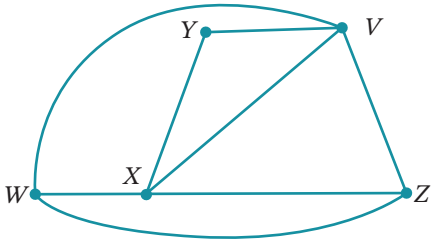
B *GHPKMPG*

C *PKMGHKM*

D *KHGPHKMG*

16 Which of the following is a cycle?

Select **A**, **B**, **C** or **D**. R C



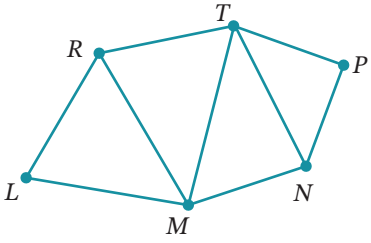
A *ZVXYVZW*

B *ZXYVXZW*

C *YXZWVY*

D *ZXVWY*

17 For the network diagram shown, write down: R C



a a trail that includes the edge *MN*

b a walk that includes the edges *PT* and *RM*

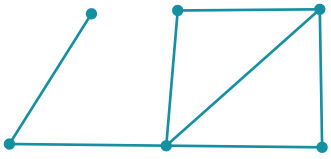
c a circuit that begins at *L* and includes edge *NT*

d a walk that begins at *T* and includes the edge *RM*

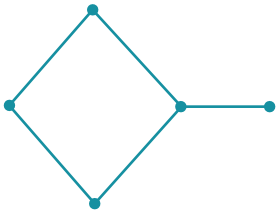
18 Count the number of faces in each network.

EXAMPLE 8

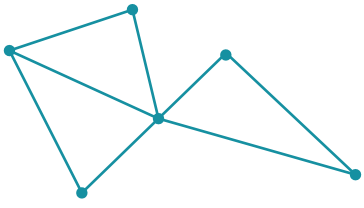
a



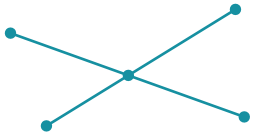
b



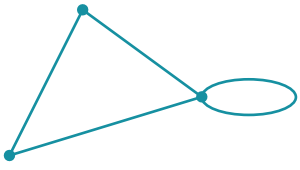
c



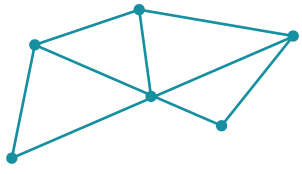
d



e



f



Foundation Standard Complex

PATH

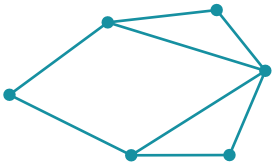
Investigation

Euler's rule

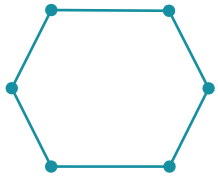
Copy and complete this table using each network below.

	Number of vertices, v	Number of faces, f	$v + f$	Number of edges, e
1				
2				
3				
4				
5				
6				
7				
8				
9				

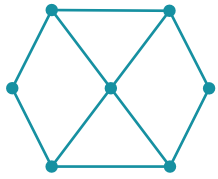
1



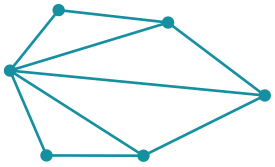
2



3



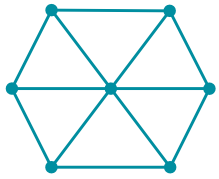
4



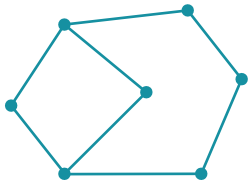
5



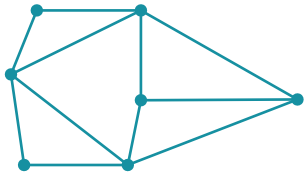
6



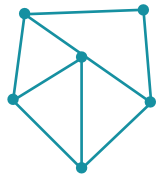
7



8



9



In 1758, the Swiss mathematician Leonhard Euler (pronounced 'oiler') discovered a relationship between $v + f$ (number of vertices and faces) and e (number of edges) for these networks.

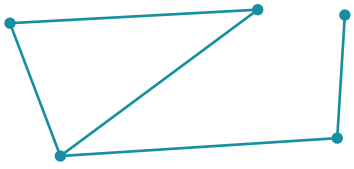
What was Euler's rule?

9780170479370

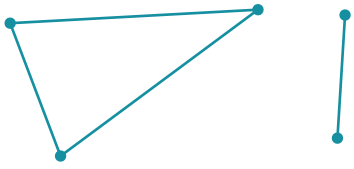
Planar graphs and Euler's formula

6.04

A network or graph is **connected** if there is a path between any 2 vertices on the network.

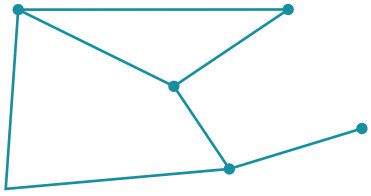


Connected

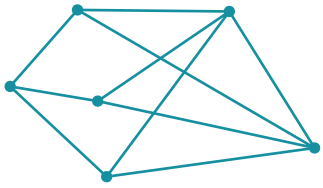


Not connected

A **planar graph** is a connected graph that can be drawn so that no edges cross over. 'Planar' means 2D, on a flat plane, and a planar graph can be drawn on a flat plane so that its edges don't overlap.

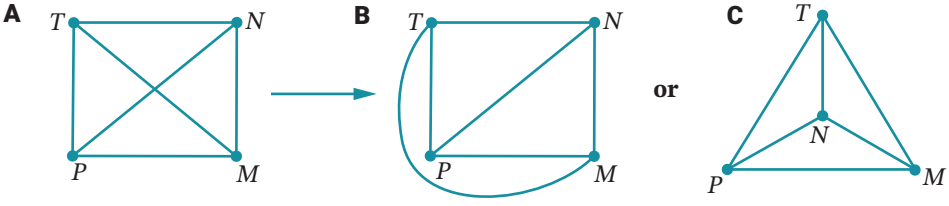


Planar graph



A non-planar graph

Some graphs may have edges crossing but they are still planar graphs because they can be *redrawn* without any of the edges crossing. For example, Graph **A** below is planar because it can be redrawn as **B** or **C**.

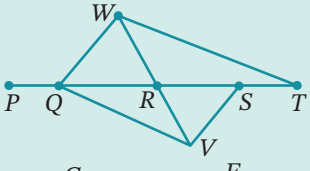


A **non-planar graph** has edges crossing and cannot be drawn so that the edges do not cross.

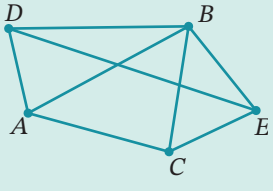
Example 9

State whether each network is a planar or non-planar graph.

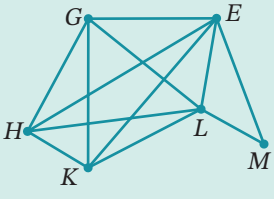
a



b



c



PATH

Video
Euler's formula for planar graphs

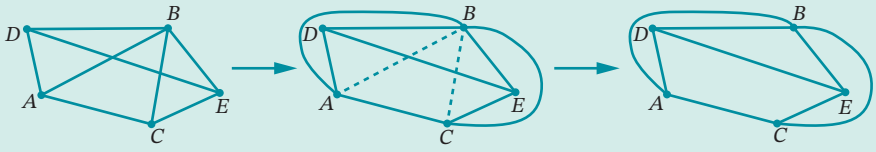
6.04

PATH

Solution

a A planar graph because no edges cross.

b The graph can be redrawn by moving AB and BC as shown.

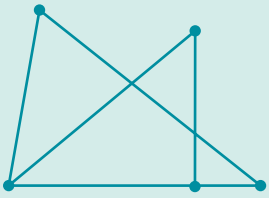


There are now no edges crossing so it's a planar graph.

c It is impossible to move the edges so that no edges cross over, so this is a non-planar graph.

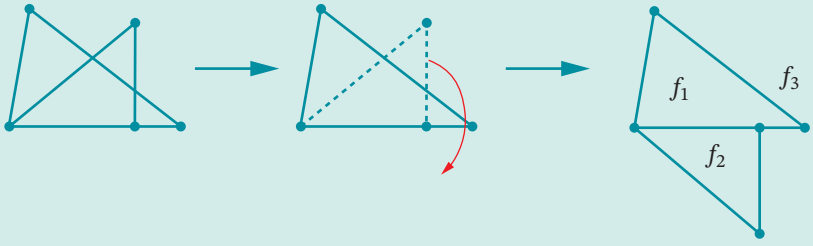
Example 10

Count the number of faces in this planar graph.



Solution

First redraw the graph so that no edges cross.



3 faces.

Euler's formula

The Swiss mathematician Leonhard Euler (pronounced 'oiler') discovered a relationship between the vertices, edges and faces of a planar graph.

Euler's formula for planar graphs

For a graph to be planar,

$$v - e + f = 2$$

where

v = number of vertices

e = number of edges

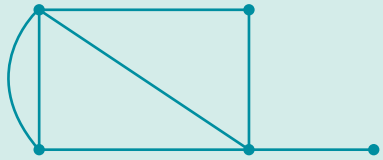
f = number of faces

Example 11

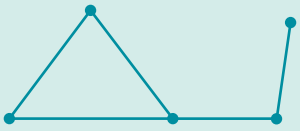
'Verify' means to show that something is true or correct.

Verify Euler's formula for each planar graph.

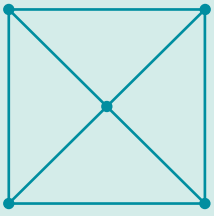
a



b



c



d



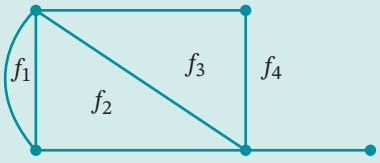
Solution

a $v = 5, e = 7$ and $f = 4$

$\therefore v - e + f = 5 - 7 + 4$

$= 2$

$\therefore$ Euler's formula is true.

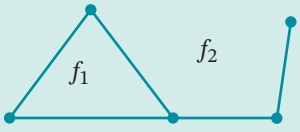


b $v = 5, e = 5$ and $f = 2$

$\therefore v - e + f = 5 - 5 + 2$

$= 2$

$\therefore$ Euler's formula is true.

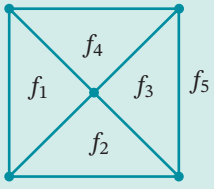


c $v = 5, e = 8$ and $f = 5$

$\therefore v - e + f = 5 - 8 + 5$

$= 2$

$\therefore$ Euler's formula is true.

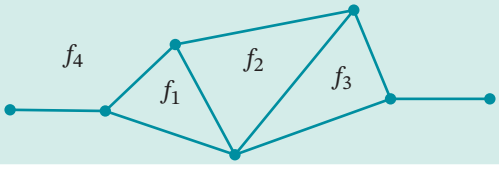


d $f = 4, v = 7$ and $e = 9$

$\therefore v - e + f = 7 - 9 + 4$

$= 2$

$\therefore$ Euler's formula is true.

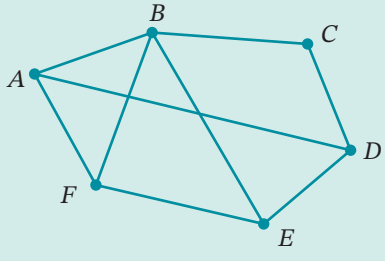


Example 12

Show that this graph is planar by:

a redrawing it so that no edges cross

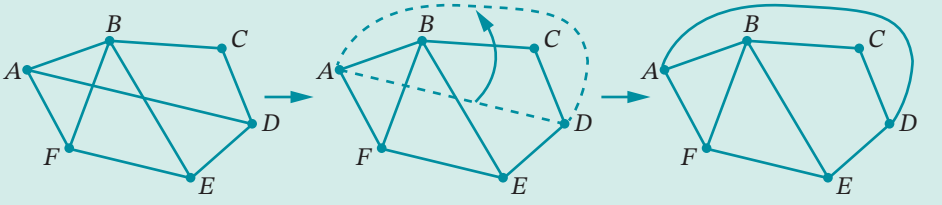
b using Euler's formula



PATH

Solution

a Edge AD crosses edges BF and BE, so move edge AD to the outside of vertices B and C.
Since the graph can be redrawn without any edges crossing, it is a planar graph.

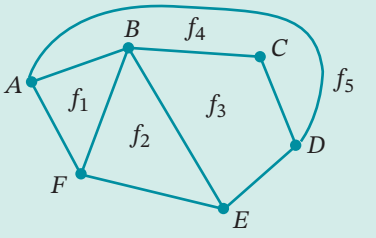


b $v = 6$, $e = 9$ and $f = 5$

$\therefore v - e + f = 6 - 9 + 5$

$= 2$

$\therefore$ The graph is planar.

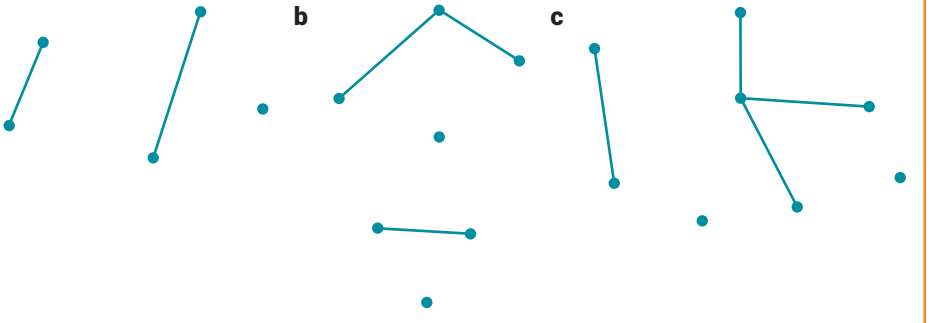


EXERCISE 6.04 ANSWERS ON P. 577

Planar graphs and Euler's formula U F R C

1 What is the smallest number of edges required to make each graph a connected network? R C

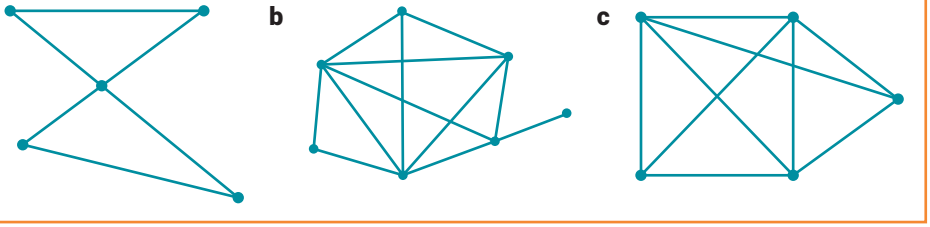
a **b** **c**



EXAMPLE 9

2 State whether each network is a planar or non-planar graph. Give a reason for your answer. C

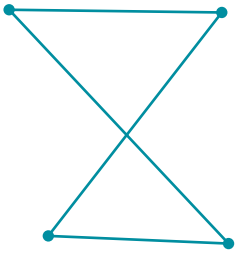
a **b** **c**

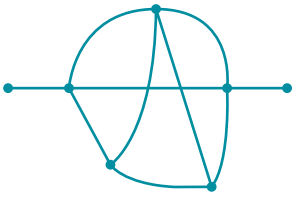


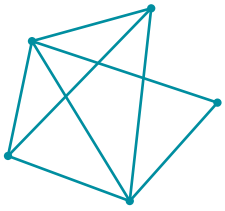
Foundation Standard Complex

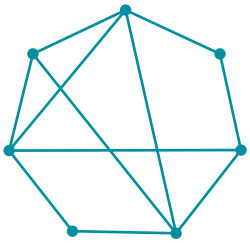
9780170479370

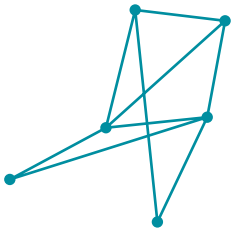
3 Redraw each graph to show that it is planar. C

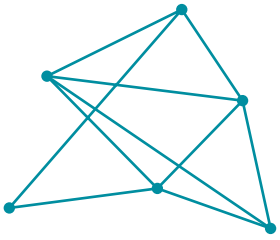
a 

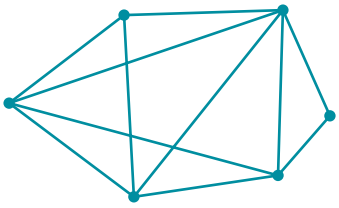
b 

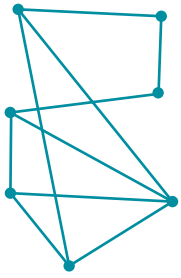
c 

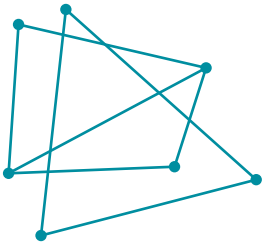
d 

e 

f 

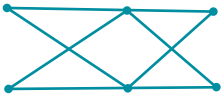
g 

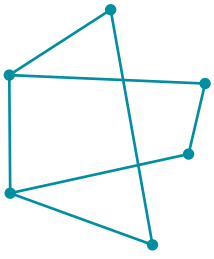
h 

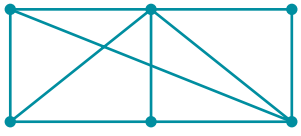
i 

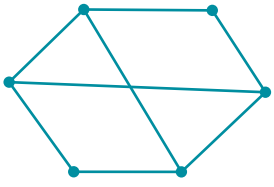
4 Redraw each planar graph to count its number of faces. R C

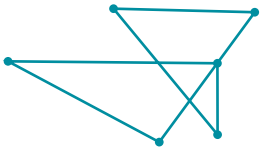
EXAMPLE 10

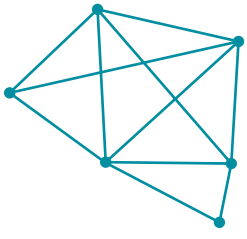
a 

b 

c 

d 

e 

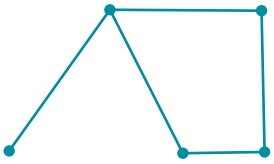
f 

PATH

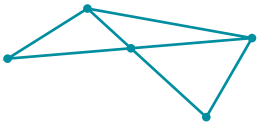
EXAMPLE 11

5 Verify Euler's formula for each planar graph. R C

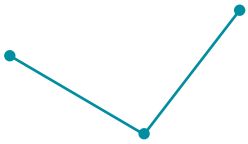
a



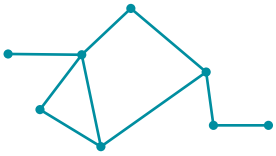
b



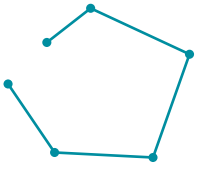
c



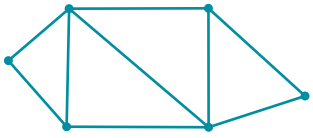
d



e

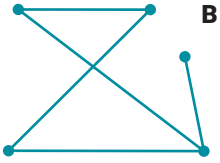


f

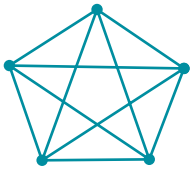


6 For which graph does Euler's formula NOT apply? R C

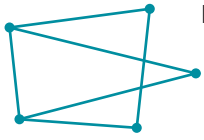
A



B



C



D

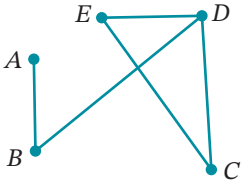


EXAMPLE 12

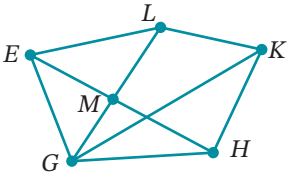
7 Show that each graph is planar by: R C

i redrawing it so that no edges cross **ii** using Euler's formula

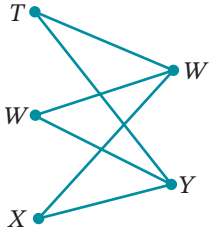
a



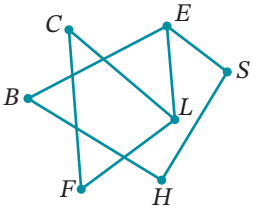
b



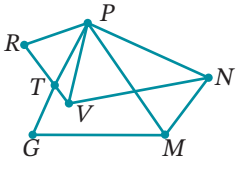
c



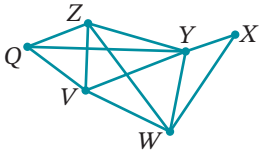
d



e



f



8 A planar graph has 14 edges and 8 faces. How many vertices has it?

Select the correct answer **A**, **B**, **C** or **D**.

A 5 **B** 6 **C** 7 **D** 8

9 A planar graph has 6 edges and 5 vertices. What is the number of faces?

Select **A**, **B**, **C** or **D**.

A 1 **B** 3 **C** 5 **D** 7

Foundation Standard Complex

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10 A planar graph has 12 vertices and 7 faces. What is the number of edges?

Select **A**, **B**, **C** or **D**.

A 20 **B** 17 **C** 14 **D** 11

11 A planar graph has 9 faces. Which of the following could be its number of vertices and edges?

Select **A, B**, **C** or **D**.

A 6 vertices and 12 edges **B** 10 vertices and 7 edges

C 9 vertices and 13 edges **D** 11 vertices and 18 edges

Power plus ANSWERS ON P. 578

1 How many diagonals has:

a a quadrilateral? **b** an octagon? **c** a dodecagon?

2 Name all quadrilaterals whose diagonals:

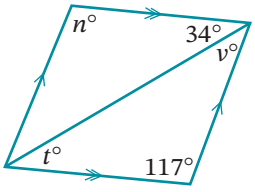
a bisect each other at right angles **b** bisect each other

c intersect at right angles **d** have equal length

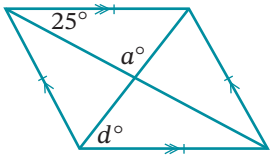
e bisect the angles of the quadrilateral **f** are equal and bisect each other

3 Find the value of each variable, giving reasons.

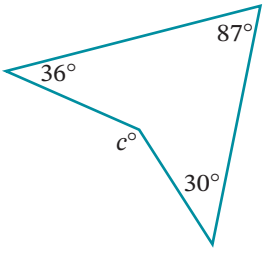
a



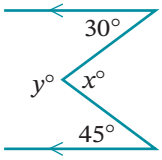
b



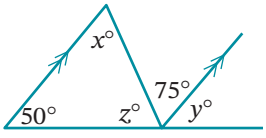
c



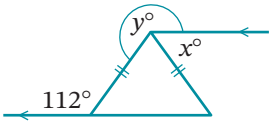
d



e



f

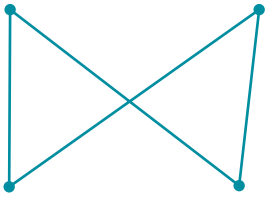


Foundation Standard Complex

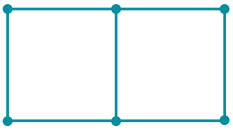
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4 Draw each network without lifting your pen or pencil (drawing each edge only once). Check your work with that of other students.

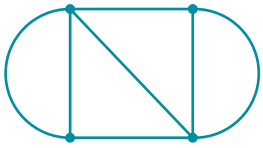
a



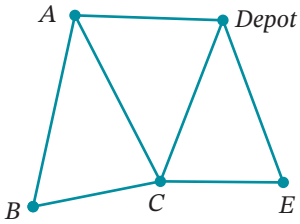
b



c



5 A roadworks crew is line marking along some roads as shown in the network diagram. The crew starts at the depot and wants to go along each road only once.



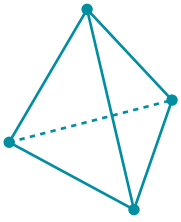
a Is it possible for the crew to go along each road only once?

b Is it possible for the crew to start and end at the depot?

6 **a** Redraw each solid as a (flat) network diagram.

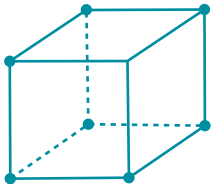
i tetrahedron

a



ii cube

b



b Are the networks of the solids planar graphs?

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CHAPTER 6 REVIEW

Quiz
Language of maths 6

Language of maths

angle sum	circuit	convex	cycle
degree	edge	equilateral	Euler
exterior angle	face	graph	isosceles
kite	network	parallelogram	path
planar graph	quadrilateral	rectangle	rhombus
square	trail	trapezium	vertex (node)

1 What shape has 2 pairs of equal adjacent sides?

2 Name one property of a convex quadrilateral.

3 What type of angle is created when one side of a triangle is extended?

4 What is the degree of a vertex in a network?

5 What is a closed walk where no vertices or edges are repeated called?

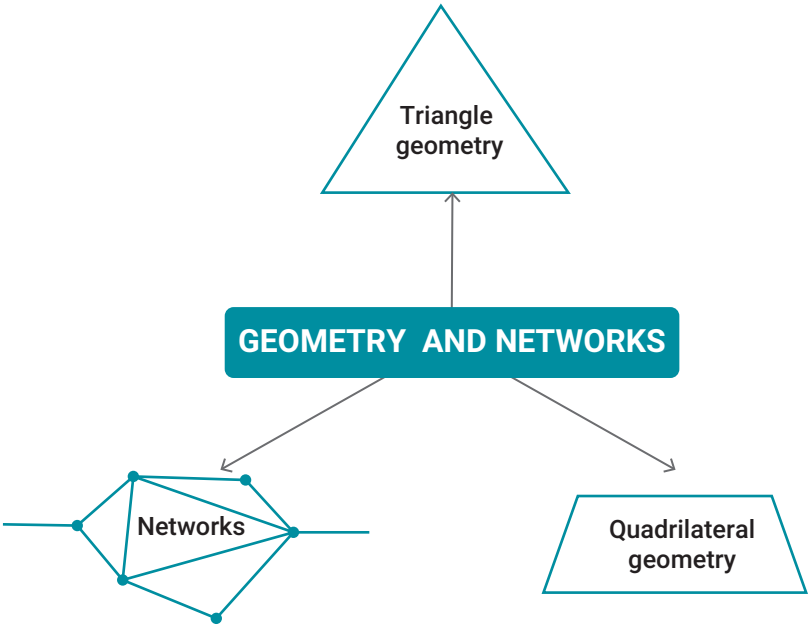
6 Write Euler's formula for planar graphs.

Topic summary

- Write 3 ideas from this topic that were new to you.
- Name the types of triangles, choose one and list all of its properties.
- Name the 6 special quadrilaterals, choose one and list all of its properties.
- Summarise what you know about the features of a network.

Print (or copy) and complete this mind map of the topic, adding detail to its branches and using pictures, symbols and colour where needed. Ask your teacher to check your work.

Worksheet
Mind map: Geometry and networks



Quiz
Test yourself 6

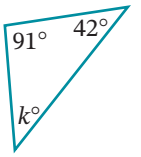
TEST YOURSELF 6

ANSWERS ON P. 578

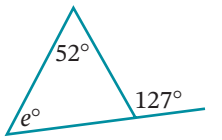
6.01

1 Find the value of each variable, giving reasons.

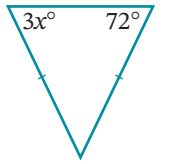
a



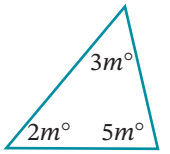
b



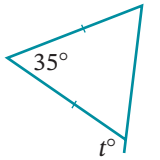
c



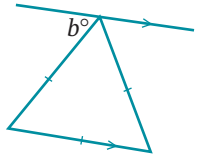
d



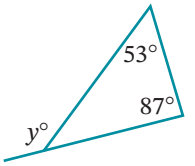
e



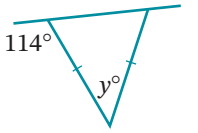
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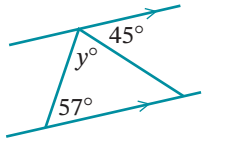
g



h



i



6.01

2 What are the sizes of the angles in:

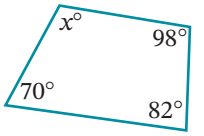
a an equilateral triangle?

b a right-angled isosceles triangle?

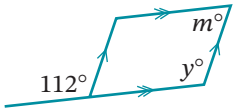
6.02

3 Find the value of each variable, giving reasons.

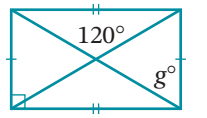
a



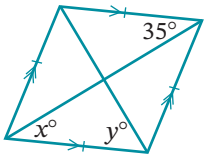
b



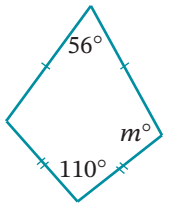
c



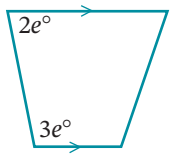
d



e



f



6.02

4 Name all quadrilaterals that have:

a both pairs of opposite sides parallel

b 2 equal diagonals

c all sides equal

d diagonals that bisect each other.

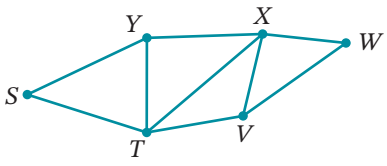
Foundation | Standard | Complex

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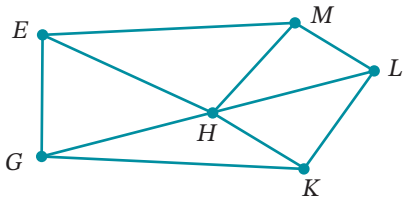
PATH

5 Find the degree of each vertex and show that the sum of the degrees is equal to twice the number of edges. 6.03

a



b



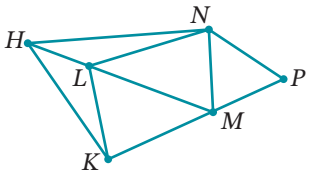
6 **a** State whether each walk is a trail or a path. 6.03

i *HLNPML*

ii *HKMPNLH*

iii *HKMNP*

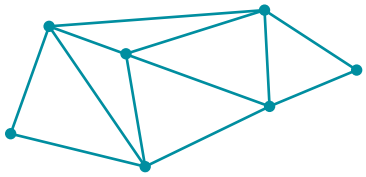
iv *LNMLKM*



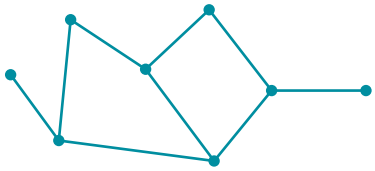
b Which of the walks in part **a** is a cycle?

7 Verify Euler's formula for each planar graph. 6.04

a



b

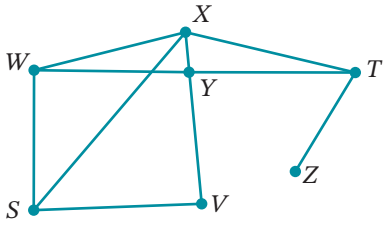


8 Show that each graph is planar by: 6.04

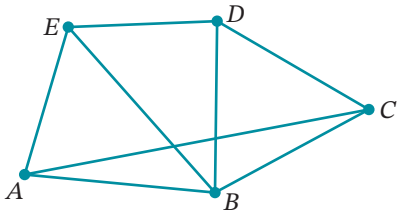
i redrawing it so that no edges cross

ii using Euler's formula

a



b



Foundation | Standard | Complex